Plot plan, elevations and isometrics

SO3 Converter Ver 1 · SO3-V1-PI-LAY-001 · Rev 0

Department of Chemical Engineering, Chulalongkorn University (teaching)

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Generated for teaching from the SO3 Converter Ver 1 operator training simulator. Every number comes from the simulator model (design point, noise off) or is marked assumed. Not for construction. No client, contractor or vendor is represented.

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1 · Purpose and scope

This document places every item of the equipment list SO3-V1-ME-EQL-001 of the SO3 Converter Ver 1 converter section on a plot plan at scale, sets the elevations, platforms and reading levels, gives five piping isometrics with bills of materials, and writes the 3-D model input layout.json with the outside-operator round. Line numbers, valves and instruments come from the EPC P&IDs SO3-PID-001/002 rev A (read from their build sources), the line list in SO3-V1-PE-HMB-001 and the instrument list SO3-V1-IN-INL-001; process data from the simulator model at its design point. Spacing rules, elevations of items not fixed by those documents, the wind and the stack height are typical practice and marked assumed.

Plant grid in metres: x east, y north, origin at the south-west fence corner; site 250 × 200 m; elevations above grade (EL 0.0; plant datum EL 100.000 m assumed). Prevailing wind assumed from the south-west (225°). Checks: 23 of 27 pass; 5 engineering queries (section 8).

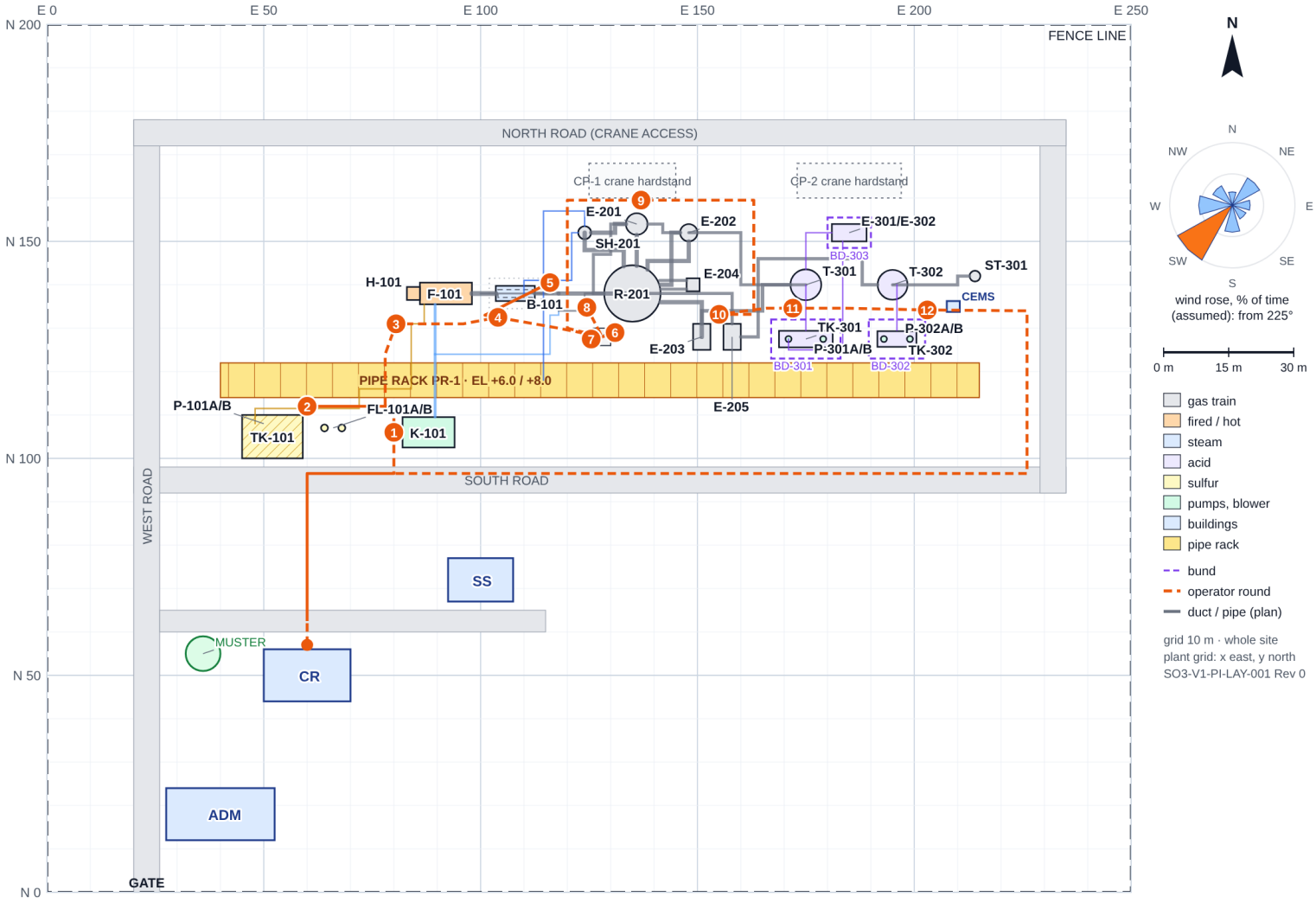
2 · Layout basis: spacing rules (typical practice, assumed)

Rule	Value used	Basis	Check	Result
Control room at least 60 m from any SO_2/SO_3 gas equipment (edge to edge)	60	typical toxic-gas spacing (CCPS siting guidance, insurer tables), assumed	R1	✓ 81.1 m
Control room at least 30 m from the acid bunds and the sulfur pit	30	typical, assumed	R2	✓ acid bunds 117.9 m, sulfur pit 44.0 m
Sulfur pit at least 15 m from the furnace and its burner	15	typical: sulfur handling apart from fired equipment, assumed	R3	✓ 35.8 m
Acid tanks inside a bund holding ≥ 110 % of the largest tank	1.1	typical containment practice, assumed	R4a	✓ 16 × 9 × 1.3 m = 187 − 22 (tank below wall top) = 166 m³
Control room upwind of the gas train: bearing within ±45° of the prevailing wind	45	prevailing wind assumed from the SW	R5	✓ control room at 224° from the gas-train centre (1° off)
Substation at least 30 m from SO_2 gas equipment and the sulfur pit, and nearer the process than the control room	30	process → utilities → buildings, typical, assumed	R6	✓ 40.6 m (control room 81.1 m)
Pumps within 10 m of the pipe-rack edge (pump rows beside the rack)	10	typical, assumed	R7	✓ P-101A/B 5.4 m, P-301A/B 4.8 m, P-302A/B 4.8 m
Every item within 40 m of a road (fire-fighting and escape)	40	typical hose reach, assumed	R8	✓ farthest H-101 32.5 m, then B-101 32.3 m
A crane hardstand within 25 m of R-201, T-301 and T-302 (catalyst and packing)	25	typical crane reach, assumed	R9	✓ R-201 15.5 m, T-301 16.5 m, T-302 16.6 m
No footprint overlap: ≥ 1.0 m clear between items (mounted items excepted), none on a road	1	typical access, assumed	R10	✓ none (22 items, 231 pairs)
Every platform above 6 m has two ways down (stair + ladder)	2	typical escape practice, assumed	R11	✗ one way down: PL-T1 (+20.2), PL-T2 (+20.2), PL-S (+30.0)
Stack at least the GEP height of the nearby structures	1	GEP rule of thumb, assumed	R12	✓ ST-301 60 m

Arrangement: the process train runs west to east north of the main E–W pipe rack PR-1 (EL +6.0 / +8.0, bents every 6 m) as on PFD-201: F-101 with H-101 on its front, B-101 with its steam drum on a structure, R-201 ringed by SH-201, E-201, E-202 and E-204, then E-203 and E-205, the absorbers T-301 and T-302 with their bunded pump tanks beside the rack, and ST-301 at the downwind (north-east) end. South of the rack: the sulfur pit TK-101 with P-101A/B and FL-101A/B next to the south road (truck unloading), and the K-101 blower house. Utilities (substation) lie between the process and the buildings; the control room, muster point and administration sit in the south-west corner, upwind. Crane hardstands CP-1 and CP-2 north of R-201 and of the absorbers serve catalyst screening and packing changes. Big gas ducts (DN2000–3400) run on their own supports, not on the rack.

3 · Plot plan

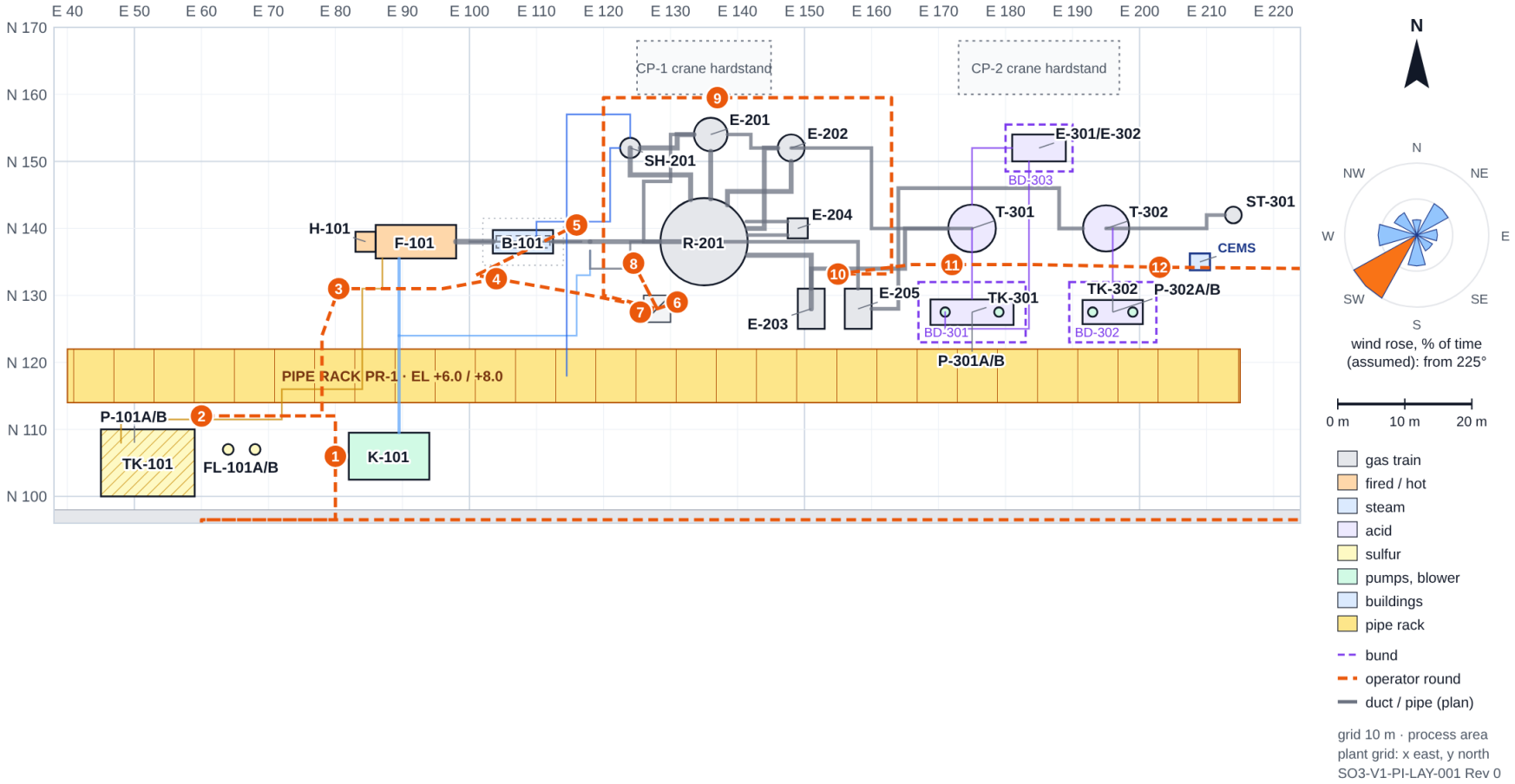
Whole site, with the outside-operator round (orange) and the pipe and duct routes in plan.



- gas train
- fired / hot
- steam
- acid
- sulfur
- pumps, blower
- buildings
- pipe rack
- bund
- operator round
- duct / pipe (plan)

grid 10 m · whole site
plant grid: x east, y north
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3.1 · Process area



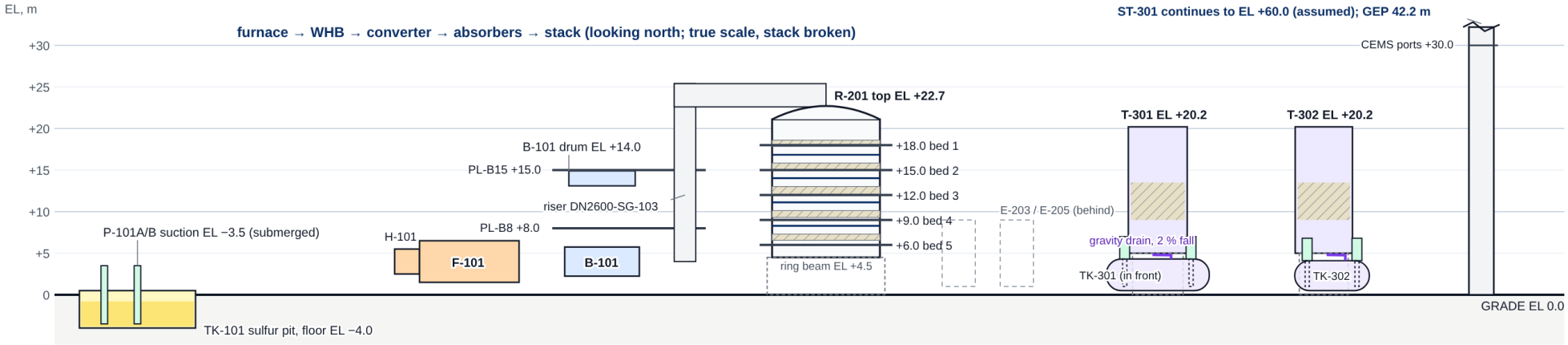
3.2 · Coordinates (centre of each item)

Tag	Service	Shape / size	x (E), m	y (N), m	Bottom EL	Top EL	Platforms	Layout note
K-101	Main air blower	12.0 × 7.0 m	88.0	106.0	0.0	+8.0	—	blower house; dried air from the drying tower (outside battery limits)
F-101	Sulfur furnace	Ø 5.0 × 12.0 m horizontal	92.0	138.0	+1.5	+6.5	—	on saddles, axis EL +4.0; burner front to the west
H-101	Preheat burner (start-up only)	3.0 × 3.0 m	84.5	138.0	+2.5	+5.5	—	start-up burner on the F-101 front
B-101	Waste heat boiler	Ø 3.5 × 9.0 m horizontal	108.0	138.0	+2.2	+5.8	+8.0, +15.0	gas axis EL +4.0 in line with F-101; steam drum above on a structure
R-201	SO_2 converter, 5 adiabatic beds (3+2)	Ø 13.0 m	135.0	138.0	+4.5	+22.7	+6.0, +9.0, +12.0, +15.0, +18.0, +22.7	on a ring beam and columns; 5 beds; platforms each bed
SH-201	Steam superheater (bed 1 gas)	Ø 3.0 m	124.0	152.0	+2.0	+10.0	—	vertical; gas in shell, steam in tubes
E-201	Hot interpass exchanger (gas–gas)	Ø 5.0 m	136.0	154.0	+2.0	+16.0	+6.0	vertical shell-and-tube (3570 m², shell ≈ 5 m assumed)
E-202	Cold interpass exchanger (gas–gas)	Ø 4.0 m	148.0	152.0	+2.0	+14.0	—	vertical shell-and-tube (1120 m², shell ≈ 4 m assumed)
E-203	Economizer before IPA	4.0 × 6.0 m	151.0	128.0	+1.0	+9.0	+6.0	finned-tube economizer casing
E-204	Interbed economizer (bed 4 → 5)	3.0 × 3.0 m	149.0	140.0	+5.0	+10.0	—	on a structure between the bed 4 outlet and bed 5 inlet nozzles
E-205	Final economizer before FAT	4.0 × 6.0 m	158.0	128.0	+1.0	+9.0	—	finned-tube economizer casing
T-301	Intermediate absorber (IPA)	Ø 7.1 m	175.0	140.0	+5.0	+20.2	+20.2	skirt to EL +5.0 so the acid drains by gravity to TK-301; T/T 14 m (assumed)
T-302	Final absorber (FAT)	Ø 6.9 m	195.0	140.0	+5.0	+20.2	+20.2	skirt to EL +5.0, gravity drain to TK-302; T/T 14 m (assumed)
TK-301	IPA pump tank	Ø 3.8 × 12.4 m horizontal	175.0	127.5	+0.5	+4.3	—	140 m³ = π/4 × 3.8² × 12.4 m
TK-302	FAT pump tank	Ø 3.6 × 9.0 m horizontal	196.0	127.5	+0.5	+4.1	—	90 m³ ≈ π/4 × 3.6² × 9 m
P-101A/B	Molten sulfur pumps (A run, B spare by hand)	Ø 1.2 m × 2	50.0	108.0	+0.5	+3.5	—	vertical submerged pumps on the pit roof; suction EL −3.5 (submerged)
FL-101A/B	Molten sulfur filters (tag assumed, not on the P&ID)	Ø 1.6 m × 2	66.0	107.0	+0.3	+3.0	—	duty/standby leaf filters beside the pit
TK-101	Molten sulfur pit (outside battery limit)	14.0 × 10.0 m pit	52.0	105.0	−4.0	+0.5	—	concrete pit 14 × 10 × 4.0 m deep (560 m³ gross for 550 m³)
P-301A/B	IPA acid pumps (B auto-start HS-301B)	Ø 1.4 m × 2	175.0	127.5	+4.3	+7.0	—	vertical cantilever pumps on the TK-301 roof
P-302A/B	FAT acid pumps (B manual)	Ø 1.4 m × 2	196.0	127.5	+4.1	+6.8	—	vertical cantilever pumps on the TK-302 roof
E-301/E-302	IPA / FAT acid coolers (acid system package, tags assumed)	8.0 × 4.0 m	185.0	152.0	+0.3	+3.5	—	two plate coolers in a small bund
ST-301	Stack	Ø 2.5 m	214.0	142.0	0.0	+60.0	+30.0	self-supporting, lined; height 60 m assumed (part 3)

3.3 · Buildings, bunds and crane hardstands

Building / area	Centre E, N	Size, m	Note
CR · Control room	60, 50	20 × 12 × 5	occupied 24 h: 6 people (assumed); blast/toxic-rated, upwind
SS · Substation	100, 72	15 × 10 × 6	unmanned; MCC for K-101 (4.2 MW) and the pumps
ADM · Administration and gate	40, 18	25 × 12 × 8	day shift: 30 people (assumed)
CEMS · Stack analyzer shelter (AIT-306/307)	209, 135	3 × 2.5 × 3	analyzer shelter at the stack base (assumed)
BD-301 · TK-301 bund	175, 127.5	16 × 9 × wall 1.3	holds TK-301, P-301A/B
BD-302 · TK-302 bund	196, 127.5	13 × 9 × wall 1.2	holds TK-302, P-302A/B
BD-303 · Acid cooler bund	185, 152	10 × 7 × wall 0.6	holds E-301/E-302
CP-1 · Crane hardstand R-201 / exchangers	135, 164	20 × 8	serves R-201, E-201, SH-201, E-202
CP-2 · Crane hardstand absorbers	185, 164	24 × 8	serves T-301, T-302, E-301/E-302

4 · Elevations and levels



R-201 stands on a ring beam at EL +4.5 so the DN2100 bed 5 outlet can leave from below; top tangent EL +21.1, top EL +22.7. One ring platform per bed manway, reached by the stair tower STR-1 and a second ladder LD-R. Bed levels (top / bottom, EL): bed 1 18.60 / 17.84; bed 2 15.84 / 15.03; bed 3 13.03 / 12.11; bed 4 10.11 / 9.30; bed 5 7.30 / 6.54.

4.1 · Levels: grade and platforms

EL, m	What is there
0.0	grade: pumps, analyzers, flow transmitters, valve stations
+0.5	sulfur pit roof (P-101A/B)
+4.1	TK-302 roof (P-302A/B discharge valves)
+4.3	TK-301 roof (P-301A/B discharge valves)
+6.0	PL-R5: bed 5 manway and thermocouples
+6.0	PL-X6: TV-231, TV-211, TV-221 bypass valves
+6.0	PL-E6: TV-251, TV-241, TE-251
+8.0	PL-B8: TV-201 bypass valve
+9.0	PL-R4: bed 4 manway and thermocouples
+9.0	PL-R9X: extension to HV-112
+12.0	PL-R3: bed 3 manway and thermocouples
+15.0	PL-R2: bed 2 manway and thermocouples
+15.0	PL-B15: drum, PSV-101A/B, FT-108, riser analyzers AT-104/105
+18.0	PL-R1: bed 1 manway and thermocouples
+20.2	PL-T1: T-301 top: TE-305, mist eliminators
+20.2	PL-T2: T-302 top: mist eliminators
+22.7	PL-RT: top head: bed 1 inlet nozzle, PT-200
+30.0	PL-S: stack sampling ports (CEMS probes)

4.2 · NPSH: pumps and submergence

NPSHa = (Patm − Pv)/(ρ g) + h_sub − h_f; typical margin NPSHa ≥ NPSHr + 1 m and ≥ 1.3 NPSHr. NPSHr is a vendor value (assumed). The acid pumps are vertical cantilever pumps in their tanks and the sulfur pumps are submerged in the pit, so the liquid level above the impeller gives the suction head.

Pump	Fluid	ρ kg/m³	Patm kPa	Pv kPa	h_sub m	h_f m	NPSHa m	NPSHr m	Margin m	OK	Note
P-301A/B	98.5 % H_2SO_4, 80 °C	1800	101.325	0.01	1.00	0.10	6.64	3.0	3.64	✓	vertical cantilever pump in TK-301; low-low level 1.0 m above the impeller eye (assumed)
P-302A/B	98.5 % H_2SO_4, 80 °C	1800	101.325	0.01	1.00	0.10	6.64	3.0	3.64	✓	in TK-302
P-101A/B	molten sulfur, 140 °C	1790	101.325	0.00	1.00	0.05	6.72	2.5	4.22	✓	vertical submerged pump in the pit; suction EL −3.5, pit low-low level 1.0 m above it (assumed)

4.3 · Gravity and self-draining lines

Line	What	From	Run m	Slope	Fall needed m	Fall drawn m	OK
L-031	T-301 acid outlet → TK-301 (gravity)	T-301 bottom EL +5.0	10.5	1:50	0.21	0.50	✓
L-033	T-302 acid outlet → TK-302 (gravity)	T-302 bottom EL +5.0	10.5	1:50	0.21	0.70	✓
DN80-MS-105	sulfur line, rack run back toward the pumps	rack x 84 → x 72	12.0	1:100	0.12	0.12	✓
DN2000-PG-301	cold wet-acid return gas: condensate falls back to T-301	E-202 end → T-301 top	39.0	1:100	0.39	0.39	✓
L-U01	saturated steam: fall with the flow to the drip leg	drum → SH-201	22.0	1:200	0.11	0.11	✓

4.4 · Stack height (worked example)

GEP rule of thumb (typical): Hg = H + 1.5 L, L = lesser of height and width of a nearby structure. Governing R-201: H = 22.7 m, L = 13.0 m → Hg = 22.7 + 1.5 × 13.0 = 42.2 m. ST-301 is set at 60 m (assumed), above GEP; ID 2.5 m at 15 m/s for 69.9 m³/s (Mechanical tab). The final height comes from a dispersion study.

Structure	H m	Width m	L m	H + 1.5 L m
R-201	22.7	13.0	13.0	42.2
T-301	20.2	7.1	7.1	30.9
T-302	20.2	6.9	6.9	30.6
E-201	16.0	5.0	5.0	23.5

5 · Isometrics

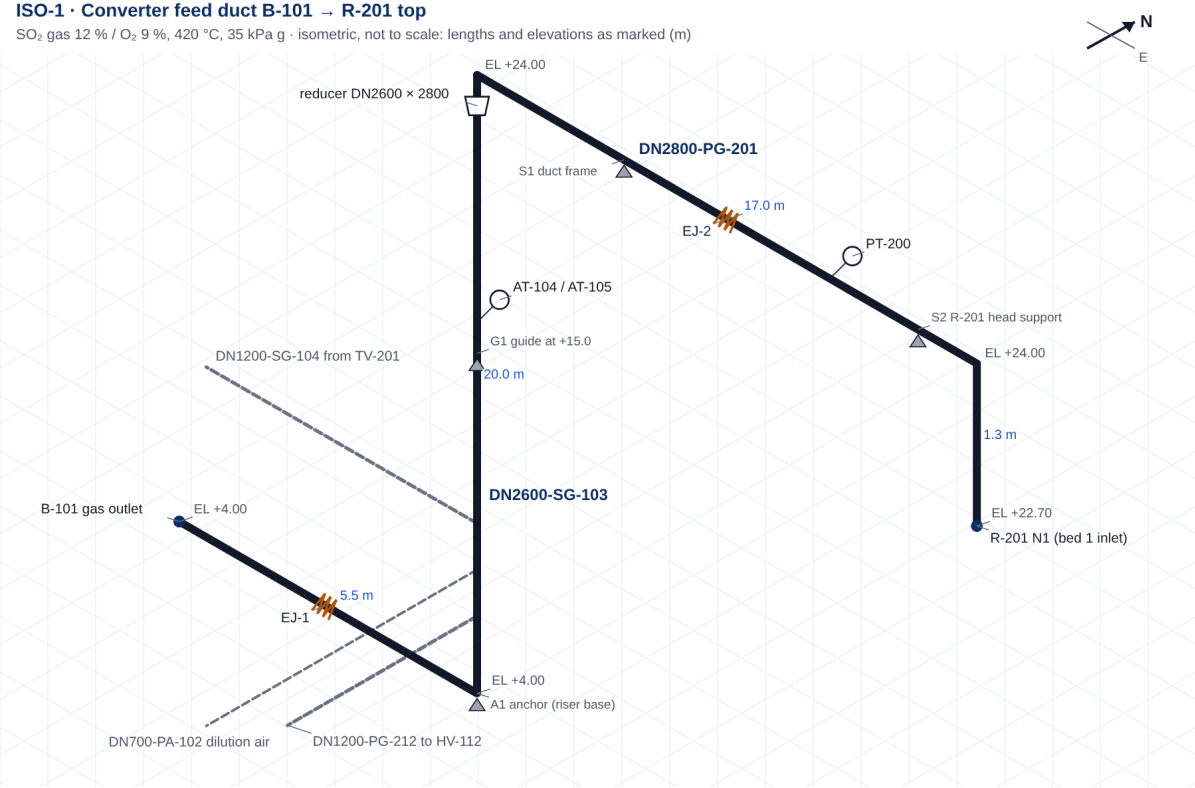
Isometric projection with north to the upper right, not to scale: long runs are shortened and every length and elevation is written on, as on a fabrication isometric. Line numbers from the EPC P&ID, or from the line list where the P&ID has none (ISO-2, ISO-4). The bill of materials is counted from the drawing.

5.1 · ISO-1 · Converter feed duct B-101 → R-201 top

Service: SO_2 gas 12 % / O_2 9 %, 420 °C, 35 kPa g. Material: carbon steel plate 10 mm, external insulation 150 mm (typical).

ISO-1 · Converter feed duct B-101 → R-201 top

SO_2 gas 12 % / O_2 9 %, 420 °C, 35 kPa g · isometric, not to scale: lengths and elevations as marked (m)



Item	Qty	Note
Pipe / duct DN700 (tie-in stub)	5 m	branch to the first flange or tee only; the branch has its own isometric
Pipe / duct DN1200 (tie-in stub)	8.5 m	branch to the first flange or tee only; the branch has its own isometric
Pipe / duct DN2600	25.5 m	carbon steel plate 10 mm, external insulation 150 mm (typical)
Pipe / duct DN2800	18.3 m	carbon steel plate 10 mm, external insulation 150 mm (typical)
Elbows 90° (mitred)	3	counted from the direction changes
Tees / branches	3	DN1200-SG-104 from TV-201; DN700-PA-102 dilution air; DN1200-PG-212 to HV-112
Reducers	1	
Valves	0	—
Expansion joints	2	
Flanges (pairs at in-line items + nozzles)	6	
Supports shown	4	typical span 12 m → ≥ 2 on 22.5 m horizontal
Instrument connections	2	AT-104 / AT-105, PT-200
Insulation / jacket surface	466 m²	outer surface at 150 mm

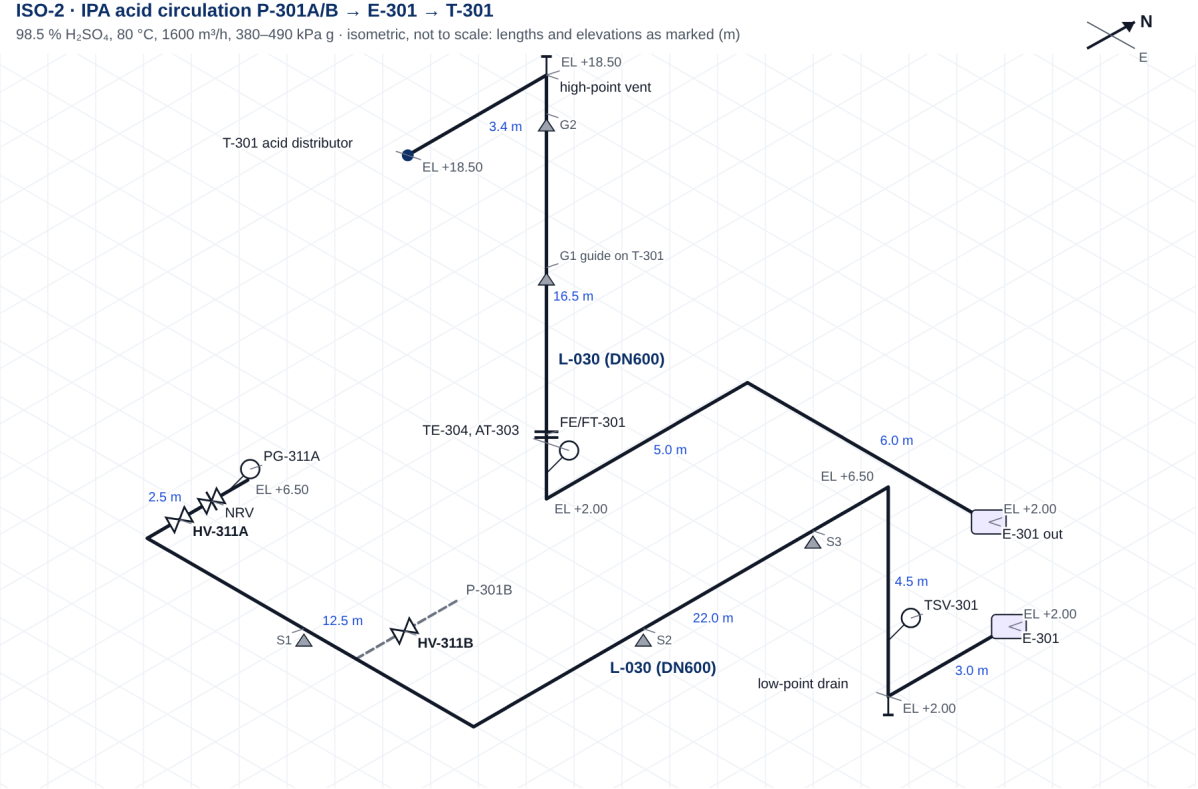
Line	Drawing	Line list	DN (line list)	DN (P&ID)	Design point T · P · velocity
DN2600-SG-103	SO3-PID-001	L-013 · Converter feed to bed 1	DN2600	DN2600	420 °C · 35.0 kPa g · 23.5 m/s
DN2800-PG-201	SO3-PID-001	L-013 · Converter feed to bed 1	DN2600	DN2800	420 °C · 35.0 kPa g · 23.5 m/s

5.2 · ISO-2 · IPA acid circulation P-301A/B → E-301 → T-301

Service: 98.5 % H_2SO_4, 80 °C, 1600 m³/h, 380–490 kPa g. Material: PTFE-lined carbon steel or high-silicon stainless (typical for 98.5 % acid).

ISO-2 · IPA acid circulation P-301A/B → E-301 → T-301

98.5 % H_2SO_4, 80 °C, 1600 m³/h, 380–490 kPa g · isometric, not to scale: lengths and elevations as marked (m)



Item	Qty	Note
Pipe / duct DN600	77.9 m	PTFE-lined carbon steel or high-silicon stainless (typical for 98.5 % acid)
Elbows 90° (mitred)	7	counted from the direction changes
Tees / branches	1	P-301B
Reducers	0	
Valves	3	NRV, HV-311A, HV-311B
Expansion joints	0	
Flanges (pairs at in-line items + nozzles)	11	
Supports shown	5	typical span 6 m → ≥ 10 on 54.4 m horizontal
Instrument connections	4	PG-311A, TSV-301, FE/FT-301, TE-304, AT-303
Insulation / jacket surface	—	acid line: no insulation, splash guards at flanges

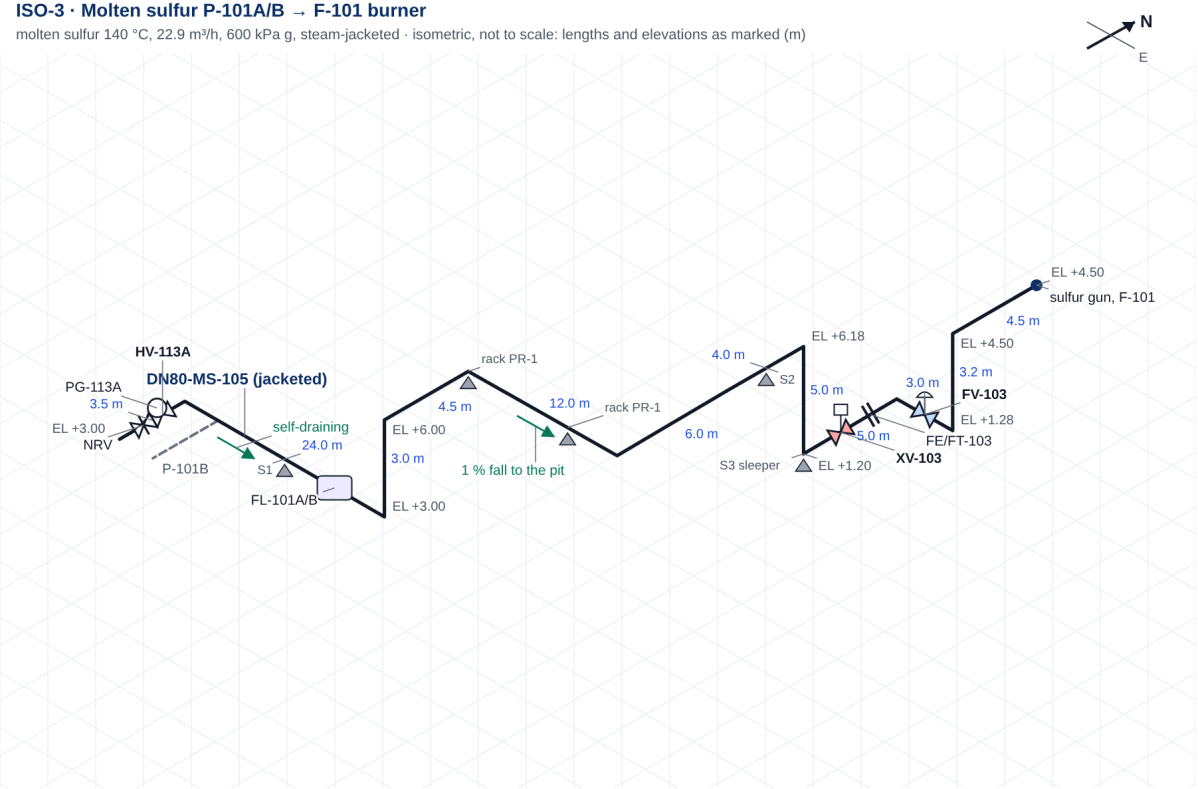
Line	Drawing	Line list	DN (line list)	DN (P&ID)	Design point T · P · velocity
L-030	line list only	L-030 · P-301 discharge to T-301	DN600	—	80 °C · 488.5 kPa g · 1.6 m/s

5.3 · ISO-3 · Molten sulfur P-101A/B → F-101 burner

Service: molten sulfur 140 °C, 22.9 m³/h, 600 kPa g, steam-jacketed. Material: carbon steel DN80 core in DN100 steam jacket, 0.4 MPa g LP steam (typical).

ISO-3 · Molten sulfur P-101A/B → F-101 burner

molten sulfur 140 °C, 22.9 m³/h, 600 kPa g, steam-jacketed · isometric, not to scale: lengths and elevations as marked (m)



Item	Qty	Note
Pipe / duct DN80	85.2 m	carbon steel DN80 core in DN100 steam jacket, 0.4 MPa g LP steam (typical)
Elbows 90° (long-radius)	10	counted from the direction changes
Tees / branches	2	P-101B
Reducers	0	
Valves	4	HV-113A, NRV, XV-103, FV-103
Expansion joints	0	
Flanges (pairs at in-line items + nozzles)	13	
Supports shown	5	typical span 3 m → ≥ 23 on 66.5 m horizontal
Instrument connections	2	PG-113A, FE/FT-103
Insulation / jacket surface	48 m²	outer surface at 50 mm

Line	Drawing	Line list	DN (line list)	DN (P&ID)	Design point T · P · velocity
DN80-MS-105	SO3-PID-001	L-006 · P-101 discharge to burner (steam-jacketed)	DN100	DN80	140 °C · 600.0 kPa g · 0.8 m/s

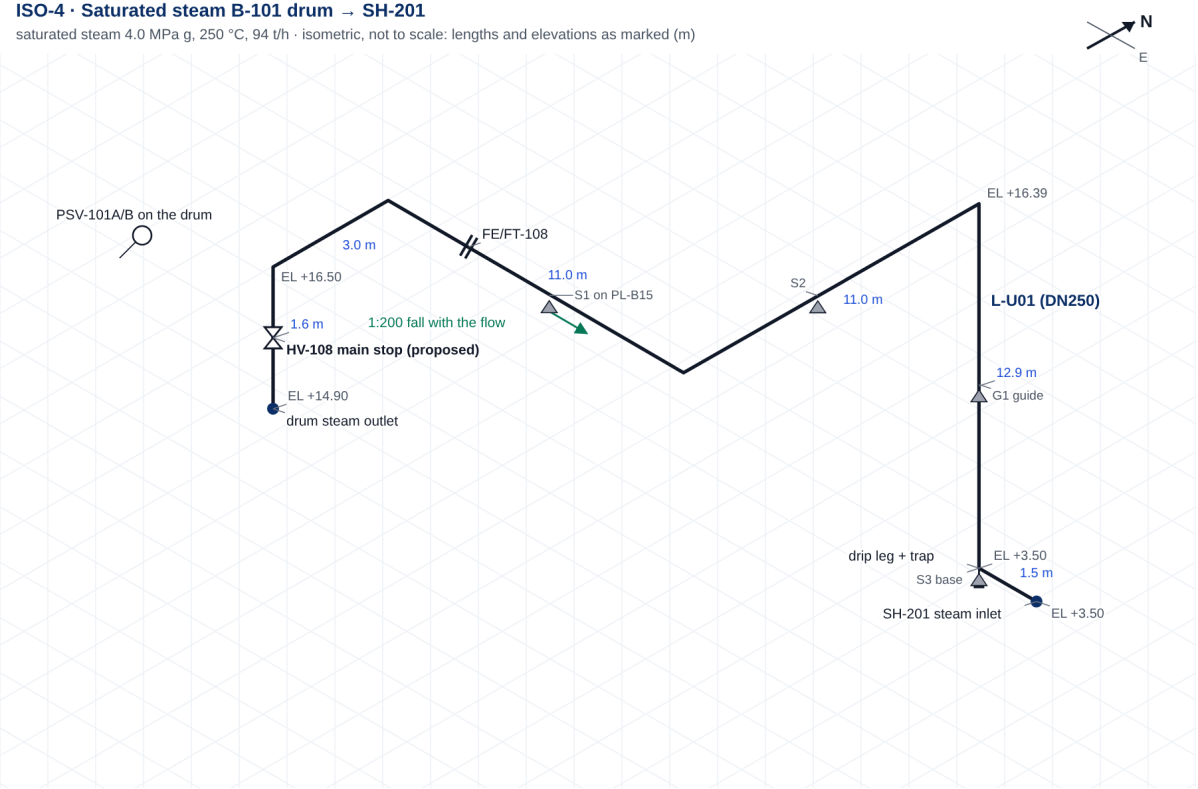
Numbers from the simulator model, design point, noise off; "assumed" where stated.

5.4 · ISO-4 · Saturated steam B-101 drum → SH-201

Service: saturated steam 4.0 MPa g, 250 °C, 94 t/h. Material: A106 Gr B, Class 600, insulation 100 mm (typical).

ISO-4 · Saturated steam B-101 drum → SH-201

saturated steam 4.0 MPa g, 250 °C, 94 t/h · isometric, not to scale: lengths and elevations as marked (m)



Item	Qty	Note
Pipe / duct DN250	41 m	A106 Gr B, Class 600, insulation 100 mm (typical)
Elbows 90° (long-radius)	5	counted from the direction changes
Tees / branches	0	—
Reducers	0	
Valves	1	HV-108 main stop (proposed)
Expansion joints	0	
Flanges (pairs at in-line items + nozzles)	6	
Supports shown	4	typical span 6.7 m → ≥ 4 on 26.5 m horizontal
Instrument connections	2	FE/FT-108, PSV-101A/B on the drum
Insulation / jacket surface	58 m²	outer surface at 100 mm

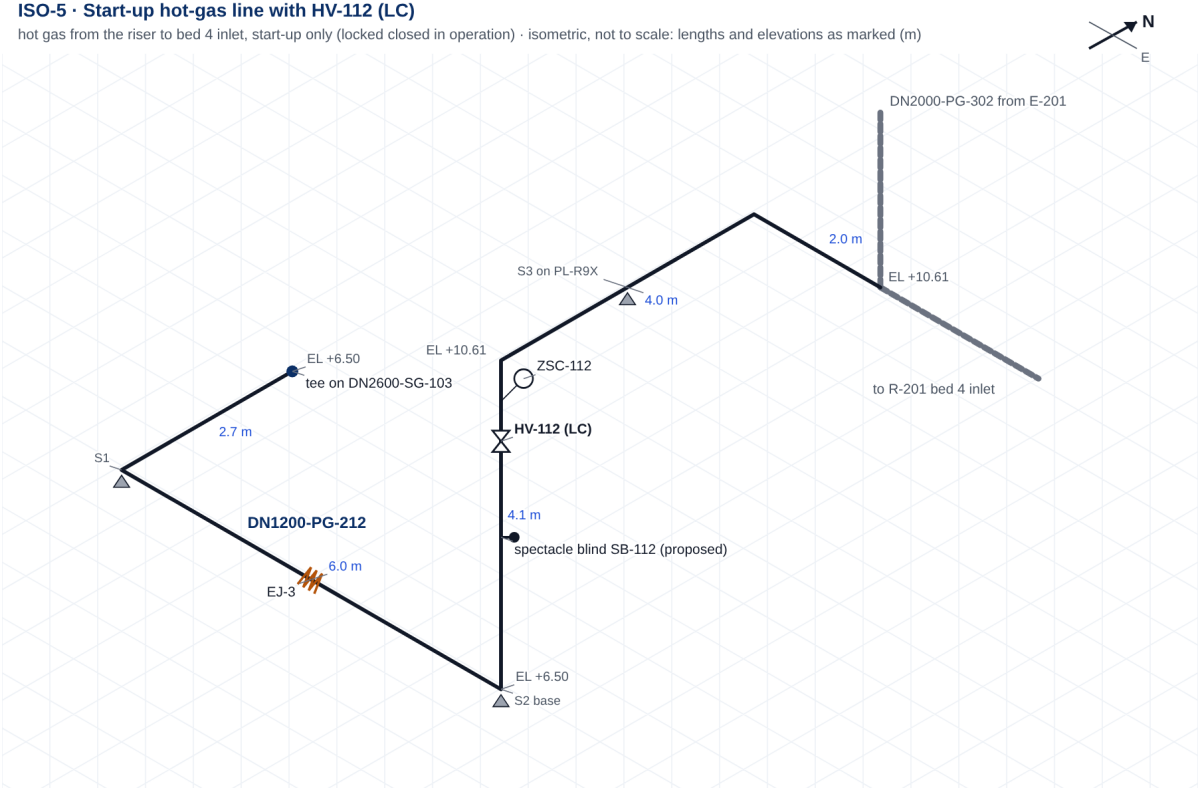
Line	Drawing	Line list	DN (line list)	DN (P&ID)	Design point T · P · velocity
L-U01	line list only	L-U01 · Saturated steam B-101 → SH-201	DN250	—	250 °C · 4000.0 kPa g · 25.7 m/s

5.5 · ISO-5 · Start-up hot-gas line with HV-112 (LC)

Service: hot gas from the riser to bed 4 inlet, start-up only (locked closed in operation). Material: 304H plate 8 mm, external insulation 150 mm (typical).

ISO-5 · Start-up hot-gas line with HV-112 (LC)

hot gas from the riser to bed 4 inlet, start-up only (locked closed in operation) · isometric, not to scale: lengths and elevations as marked (m)



Item	Qty	Note
Pipe / duct DN1200	18.8 m	304H plate 8 mm, external insulation 150 mm (typical)
Pipe / duct DN2000 (tie-in stub)	4.9 m	branch to the first flange or tee only; the branch has its own isometric
Elbows 90° (mitred)	4	counted from the direction changes
Tees / branches	2	DN2000-PG-302 from E-201; to R-201 bed 4 inlet
Reducers	0	
Valves	1	HV-112 (LC)
Expansion joints	1	
Flanges (pairs at in-line items + nozzles)	7	
Supports shown	3	typical span 12 m → ≥ 2 on 14.7 m horizontal
Instrument connections	1	ZSC-112
Insulation / jacket surface	124 m²	outer surface at 150 mm

Line	Drawing	Line list	DN (line list)	DN (P&ID)	Design point T · P · velocity
DN1200-PG-212	SO3-PID-002	—	—	DN1200	—

6 · From layout to the 3-D model: layout.json and the operator round

layout.json holds: site (fence, roads, rack with tiers, buildings with occupancy, bunds, crane hardstands, stair towers, wind), eq (boxes and cylinders with dimensions, bottom and top EL, colour class), aux (B-101 drum and structure), pipes (34 xyz polylines with DN, service and P&ID / line-list numbers), platforms (14) and ladders (11), r201 (bed and nozzle levels), tags (97: x, y, z, the level read from, what is read there, the live reading key; 10 hand valves of which 17 proposed items) and route.

The round: 12 stations in flow order from the control room and back, 751 m walk. A tag counts as reached within 20 m of the station and on one of its levels.

Station	Name	Levels EL	Camera point (x, y, z)	Tags read or operated	n
1	K-101 blower house	0.0	80.0, 106.0, 1.6	K-101, FE-100, FT-100, PT-101	4
2	Sulfur pit, pumps and filters	0.0, +0.5	60.0, 112.0, 1.6	P-101A, P-101B, HV-113A, HV-113B, PG-113A, PG-113B, HV-114A, HV-114B, PDG-114	9
3	Furnace front: H-101, sulfur and air valves	0.0	80.5, 131.0, 1.6	H-101, HV-109, FT-109, FE-101, FT-101, FV-101, XV-103, XY-103, ZSC-103, ZSO-103, FE-103, FT-103, FV-103	13
4	B-101 waste heat boiler and TV-201 platform	0.0, +8.0	104.0, 132.5, 1.6	TE/TT-106, TE/TT-107, TV-201, FE-102, FT-102, FV-102	6
5	Drum and riser platform +15.0	+15.0	116.0, 140.5, 16.6	FE-108, FT-108, PSV-101A, PSV-101B, HV-108, LG-108, AT-104, AT-105	8
6	R-201 top and bed 1 (+22.7, +18.0)	+22.7, +18.0	131.0, 129.0, 24.3	PT-200, TE/TT-201, TE/TT-202, TE/TT-203, PDT-204	5
7	R-201 beds 2 and 3 (+15.0, +12.0)	+15.0, +12.0	125.5, 127.5, 16.6	TE/TT-211, TE/TT-212, TE/TT-213, PDT-214, TE/TT-221, TE/TT-222, TE/TT-223, PDT-224	8
8	R-201 beds 4 and 5, HV-112 (+9.0, +6.0)	+9.0, +6.0	124.5, 134.8, 10.6	TE/TT-232, TE/TT-233, PDT-234, TE/TT-241, TE/TT-242, TE/TT-243, PDT-244, TE/TT-231, HV-112, ZSC-112	10
9	Interpass exchangers SH-201, E-201, E-202	0.0, +6.0	137.0, 159.5, 1.6	TV-231, TV-211, TV-221, TE/TT-207, TE/TT-253, TE/TT-254	6
10	Economizers E-203, E-204, E-205	0.0, +6.0	155.0, 133.2, 1.6	TV-251, TE/TT-251, TV-241, TE/TT-255, FT-253, TE/TT-252	6
11	T-301 and the IPA pump tank TK-301	0.0, +4.3, +20.2	172.0, 134.6, 1.6	P-301A, P-301B, HV-311A, HV-311B, PG-311A, PG-311B, FE-301, FT-301, AT-303, TE/TT-304, TE/TT-305	11
12	T-302, TK-302, acid coolers and stack base	0.0, +4.1	203.0, 134.2, 1.6	P-302A, P-302B, HV-321A, HV-321B, PG-321A, PG-321B, FE-302, FT-302, AT-306, AT-307	10

7 · Re-link: Consequence analysis on this plot plan

The Consequence analysis tab screened its scenarios on a provisional plan. Its cq_state.json is now written from layout.json (tabs/cq/tools/mk_state.js): sources at their equipment or line, targets at the control room, substation, administration building and fence. Design state, provisional → this layout:

Scenario	Source E, N	Control room m	Substation m	Administration m	Fence m	N on site	N off site	S
S1 Toxic SO_2 gas release	118, 138	65 → 99	79 → 63	143 → 137	52 → 62	3.90 → 1.23	0.50 → 0.40	5 → 5
S2 SO_3-bearing gas to the IPA	168.2, 140	77 → 135	121 → 91	176 → 171	60 → 60	2.75 → 1.19	0.50 → 0.50	5 → 5
S3 Hot strong-acid leak	171, 127.5	89 → 129	139 → 85	190 → 165	41 → 73	1.01 → 1.01	0.00 → 0.00	4 → 4
S4 Molten sulfur pool fire	52, 105	91 → 50	55 → 53	118 → 82	32 → 52	48.04 → 48.04	382.10 → 381.40	5 → 5
S5 Physical burst of B-101	108, 138	62 → 94	69 → 61	133 → 132	52 → 62	18.16 → 16.79	18.77 → 16.81	5 → 5
S6 Start-up fuel gas leak	82, 136.5	69 → 83	50 → 62	116 → 120	48 → 64	1.50 → 1.50	0.00 → 0.00	5 → 5

8 · Checks and engineering queries

No.	Group	Check	Expected	Found	Result	Query
C1	Completeness	Every item of the equipment list SO3-V1-ME-EQL-001 is on the plot plan	22 items	22 of 22	✓	
C2	Completeness	Every PFD-201 item is on the plot plan	13 items	13 of 13	✓	
C3	Completeness	Every isometric line number exists on the EPC P&ID or in the line list	6 line numbers	6 of 6 found (DN2600-SG-103 P&ID; DN2800-PG-201 P&ID; L-030 line list; DN80-MS-105 P&ID; L-U01 line list; DN1200-PG-212 P&ID)	✓	
C4	Completeness	Isometric and tie-in sizes: P&ID rev A DN = line list DN	equal	DN2800-PG-201 vs L-013 DN2600; DN80-MS-105 vs L-006 DN100; DN2000-PG-302 vs L-024 DN2700; DN1200-SG-104 vs L-009 DN1600; DN700-PA-102 vs L-004 DN800	✗	EQ-la-02
C5	Completeness	Every P&ID line number is modelled as a pipe run in layout.json, and every run number is on the P&ID	17 P&ID numbers	all 17 modelled	✓	
C5b	Completeness	Every process duct, acid and steam line in the model carries a P&ID line number	28 runs	15 without: L-007 (L-007), PG-202 (L-015), SH-E201, E202-E201, PG-303 (L-025), PG-303R, L-028 (L-028), L-030a (L-030), L-030b (L-030), L-031 (L-031), L-032 (L-032), L-032b (L-032), L-033 (L-033), L-U01 (L-U01), L-U02 (L-U02)	✗	EQ-la-04
C6	Field positions	Every P&ID hand valve, control valve and instrument has a field position in layout.json	58 P&ID tags (10 valves incl. HV-112)	58 of 58	✓	
C7	Field positions	Every physical item of the instrument list SO3-V1-IN-INL-001 has a position	75 items	75 of 75	✓	
C8	Field positions	Hand valves and local gauges on the P&ID rev A (needed for the field round)	block valve + gauge at every pump	1 on rev A (HV-112); 17 proposed here	✗	EQ-la-01
C9	Route	Route stations reach every field tag (≤ 20 m and on a station level)	96 tags	96 of 96	✓	
C10	Route	The round has about 12 stations, in flow order, starting and ending at the control room	10–14 stations, closed	12 stations, 751 m walk, closed	✓	
R1	Spacing	Control room at least 60 m from any SO_2/SO_3 gas equipment (edge to edge)	≥ 60 m	81.1 m	✓	
R2	Spacing	Control room at least 30 m from the acid bunds and the sulfur pit	≥ 30 m	acid bunds 117.9 m, sulfur pit 44.0 m	✓	
R3	Spacing	Sulfur pit at least 15 m from the furnace and its burner	≥ 15 m	35.8 m	✓	
R4a	Spacing	TK-301 inside bund BD-301 holding ≥ 110 % of the tank	≥ 154 m^3 net, tank inside	16 × 9 × 1.3 m = 187 − 22 (tank below wall top) = 166 m^3	✓	
R4b	Spacing	TK-302 inside bund BD-302 holding ≥ 110 % of the tank	≥ 99 m^3 net, tank inside	13 × 9 × 1.2 m = 140 − 13 (tank below wall top) = 128 m^3	✓	
R5	Spacing	Control room upwind of the gas train: bearing within ±45° of the prevailing wind	within ±45° of 225°	control room at 224° from the gas-train centre (1° off)	✓	
R6	Spacing	Substation at least 30 m from SO_2 gas equipment and the sulfur pit, and nearer the process than the control room	≥ 30 m and < control room	40.6 m (control room 81.1 m)	✓	
R7	Spacing	Pumps within 10 m of the pipe-rack edge (pump rows beside the rack)	≤ 10 m	P-101A/B 5.4 m, P-301A/B 4.8 m, P-302A/B 4.8 m	✓	
R8	Spacing	Every item within 40 m of a road (fire-fighting and escape)	≤ 40 m	farthest H-101 32.5 m, then B-101 32.3 m	✓	
R9	Spacing	A crane hardstand within 25 m of R-201, T-301 and T-302 (catalyst and packing)	≤ 25 m	R-201 15.5 m, T-301 16.5 m, T-302 16.6 m	✓	
R10	Spacing	No footprint overlap: ≥ 1.0 m clear between items (mounted items excepted), none on a road	no clash	none (22 items, 231 pairs)	✓	
R11	Spacing	Every platform above 6 m has two ways down (stair + ladder)	≥ 2 for 11 platforms	one way down: PL-T1 (+20.2), PL-T2 (+20.2), PL-S (+30.0)	✗	EQ-la-03
R12	Spacing	Stack at least the GEP height of the nearby structures	≥ 42.2 m (R-201)	ST-301 60 m	✓	
E1	Elevations	NPSH available ≥ NPSH required + 1 m (and ≥ 1.3 ×) for every pump	3 pumps	P-301A/B 6.6 / 3.0 m, P-302A/B 6.6 / 3.0 m, P-101A/B 6.7 / 2.5 m	✓	

Numbers from the simulator model, design point, noise off; "assumed" where stated.

No.	Group	Check	Expected	Found	Result	Query
E2	Elevations	Gravity and self-draining lines have their fall	5 lines	L-031 0.50/0.21 m, L-033 0.70/0.21 m, DN80-MS-105 0.12/0.12 m, DN2000-PG-301 0.39/0.39 m, L-U01 0.11/0.11 m	✓	
E3	Elevations	Every reading point is within reach of its standing level (|z − level| ≤ 3 m)	97 tags	all within 3 m	✓	

8.1 · Engineering queries raised by layout

No.	Document	Discipline	Subject	Finding	Proposed
EQ-la-01	SO3-PID-001/002 rev A	Piping / Process	Hand valves and local gauges for the field round	The EPC P&IDs rev A show one hand valve (HV-112, locked closed) and no local gauges. The layout, the 3-D model and the outside-operator round need the pump discharge block and check valves and local pressure gauges at P-101A/B, P-301A/B and P-302A/B, the change-over valves of FL-101A/B, the main steam stop valve and the drum gauge glass.	Show them on rev B; this document proposes 17 tags with positions in layout.json (HV-113A, HV-113B, PG-113A, PG-113B, HV-114A, HV-114B, PDG-114, HV-108, LG-108, HV-311A, HV-311B, PG-311A, PG-311B, HV-321A, HV-321B, PG-321A, PG-321B), to be renumbered by the P&ID owner if needed.
EQ-la-02	SO3-PID-001/002 rev A vs SO3-V1-PE-HMB-001	Process / Piping	Line sizes and service codes: P&ID vs line list	On the isometric and tie-in lines the P&ID rev A size differs from the line list: DN2800-PG-201 vs L-013 DN2600; DN80-MS-105 vs L-006 DN100; DN2000-PG-302 vs L-024 DN2700; DN1200-SG-104 vs L-009 DN1600; DN700-PA-102 vs L-004 DN800. Service codes also differ (P&ID MS molten sulfur vs line list SL; the converter feed is SG on the riser and PG on the top duct, one line-list line L-013).	Re-issue the P&ID line numbers with the line-list sizes (sized on the 110 % case), one service code per fluid; the second-pass ducts DN2000 vs DN2700 are the same point as EQ-em-06. The isometrics keep the P&ID numbers and show both sizes until then.
EQ-la-03	SO3-V1-PI-LAY-001	Piping / Safety	Platforms with one way down	one way down: PL-T1 (+20.2), PL-T2 (+20.2), PL-S (+30.0) have one caged ladder each. The typical rule used here asks for two ways down from every platform above 6 m.	Add a second ladder on the opposite side of the T-301 and T-302 top platforms (8 × 8 m) or reduce them to a dead end of 7.5 m or less; accept one ladder with fall arrest on the stack sampling platform (occasional access) in the safety review.
EQ-la-04	SO3-PID-001/002 rev A	Process / Piping	Lines without a P&ID line number	15 without: L-007 (L-007), PG-202 (L-015), SH-E201, E202-E201, PG-303 (L-025), PG-303R, L-028 (L-028), L-030a (L-030), L-030b (L-030), L-031 (L-031), L-032 (L-032), L-032b (L-032), L-033 (L-033), L-U01 (L-U01), L-U02 (L-U02). The bed 1 and bed 4 outlet labels are hidden on rev A (the equipment list quotes DN2900-PG-202 and DN2000-PG-303); the SH-201 → E-201, E-202 → E-201 and E-205 → T-302 ducts are unnumbered; the acid and steam lines exist only in the line list because their P&IDs are outside the OTS scope.	Show every line number on rev B and allocate numbers for the acid and steam lines on the acid-system and steam-system P&IDs; ISO-2 and ISO-4 carry the line-list numbers L-030 and L-U01 until then.
EQ-la-05	Consequence analysis tab / SO3-V1-SA-HZP-001	Safety	Consequence screening re-run on the layout	With the layout of this document the control room is 99 m from the S1 SO_2 source (provisional plan 65 m) and on-site N for S1 falls from 3.898 to 1.234; the sulfur pit (S4) is now 50 m from the control room (was 91 m). Severity is unchanged for all six scenarios (S5 S5 S4 S5 S5 S5).	No change to the HAZOP/LOPA severities; keep the control room at ≥ 60 m from the gas train and upwind (checks R1, R5) in later layout revisions and re-run the screening after any move of the control room or the sulfur pit.

Appendix A · Reading points and hand valves (layout.json tags)

Tag	Kind	What is read or operated there	x, y	EL item	Read from EL	Station	P&ID
K-101	lcs	blower local control station: run, stop, amps (assumed)	86.0, 109.9	+1.2	0.0	1	rev A / list
FE-100	fe	orifice on DN1800-PA-101	89.5, 111.0	+3.0	0.0	1	rev A / list
FT-100	tx	total air (also FSLL-100, I-1)	90.8, 111.0	+1.5	0.0	1	rev A / list
PT-101	tx	blower discharge pressure	90.8, 112.2	+1.5	0.0	1	rev A / list
P-101A	lcs	pump local control station	48.0, 109.4	+1.2	+0.5	2	rev A / list
P-101B	lcs	pump local control station	52.0, 109.4	+1.2	+0.5	2	rev A / list
HV-113A	hv	P-101A discharge block valve	48.0, 110.3	+3.0	+0.5	2	proposed
HV-113B	hv	P-101B discharge block valve	52.0, 110.3	+3.0	+0.5	2	proposed
PG-113A	pg	P-101A discharge pressure gauge	48.6, 109.8	+2.4	+0.5	2	proposed
PG-113B	pg	P-101B discharge pressure gauge	52.6, 109.8	+2.4	+0.5	2	proposed
HV-114A	hv	FL-101A inlet block valve (filter change-over)	64.0, 109.3	+2.0	0.0	2	proposed
HV-114B	hv	FL-101B inlet block valve (stand-by)	68.0, 109.3	+2.0	0.0	2	proposed
PDG-114	pg	filter differential pressure gauge	66.0, 109.3	+1.6	0.0	2	proposed
H-101	lcs	burner package panel: flame on, firing	83.5, 135.8	+1.2	0.0	3	rev A / list
HV-109	cv	H-101 firing valve (burner package)	82.0, 136.0	+2.0	0.0	3	rev A / list
FT-109	tx	H-101 fuel meter	82.0, 132.0	+1.5	0.0	3	rev A / list
FE-101	fe	orifice on DN1800-PA-103	89.5, 127.0	+3.5	0.0	3	rev A / list
FT-101	tx	combustion air	91.0, 127.0	+1.5	0.0	3	rev A / list
FV-101	cv	combustion-air valve position (FO)	89.5, 131.0	+3.0	0.0	3	rev A / list
XV-103	xv	sulfur shut-off valve (SIS, spring return)	84.0, 128.0	+1.2	0.0	3	rev A / list
XY-103	rly	XV-103 solenoid	84.4, 128.0	+1.6	0.0	3	rev A / list
ZSC-103	zs	XV-103 closed limit switch	84.4, 128.3	+1.5	0.0	3	rev A / list
ZSO-103	zs	XV-103 open limit switch	84.4, 127.7	+1.5	0.0	3	rev A / list
FE-103	fe	sulfur flow element (jacketed)	84.0, 129.6	+1.2	0.0	3	rev A / list
FT-103	tx	sulfur flow	83.0, 129.6	+1.4	0.0	3	rev A / list
FV-103	cv	sulfur flow valve position (FC)	85.5, 131.0	+1.3	0.0	3	rev A / list
TE/TT-106	tx	furnace outlet temperature	98.5, 136.2	+2.0	0.0	4	rev A / list
TE/TT-107	tx	B-101 gas outlet temperature	113.0, 136.6	+2.5	0.0	4	rev A / list
TV-201	cv	B-101 hot-gas bypass valve position (FC)	110.0, 138.0	+9.5	+8.0	4	rev A / list
FE-102	fe	orifice on DN700-PA-102	116.0, 127.0	+2.5	0.0	4	rev A / list
FT-102	tx	dilution air	117.2, 127.0	+1.5	0.0	4	rev A / list
FV-102	cv	dilution-air valve position (FO)	116.0, 130.0	+2.5	0.0	4	rev A / list
FE-108	fe	orifice on the saturated steam line	113.0, 141.0	+16.5	+15.0	5	rev A / list
FT-108	tx	steam from B-101	113.0, 140.0	+16.2	+15.0	5	rev A / list
PSV-101A	psv	drum safety valve (relief basis SO3-V1-PE-RLB-001), check seal	106.0, 138.0	+15.6	+15.0	5	rev A / list
PSV-101B	psv	drum safety valve, check seal	109.0, 138.0	+15.6	+15.0	5	rev A / list
HV-108	hv	main steam stop valve on the drum outlet	110.0, 138.4	+15.7	+15.0	5	proposed
LG-108	pg	drum gauge glass (steam system)	104.5, 139.2	+14.0	+15.0	5	proposed
AT-104	tx	SO_2 to bed 1 (probe on the riser)	119.4, 138.0	+16.0	+15.0	5	rev A / list
AT-105	tx	O_2 to bed 1 (probe on the riser)	119.4, 137.2	+16.0	+15.0	5	rev A / list
PT-200	tx	converter inlet pressure	130.0, 138.0	+24.0	+22.7	6	rev A / list

Tag	Kind	What is read or operated there	x, y	EL item	Read from EL	Station	P&ID
TE/TT-201	tx	bed 1 inlet temperature	128.3, 136.5	+19.9	+18.0	6	rev A / list
TE/TT-202	tx	bed 1 middle temperature	128.3, 138.0	+18.2	+18.0	6	rev A / list
TE/TT-203	tx	bed 1 exit temperature (also TSHH-203, I-2)	128.3, 139.5	+17.5	+18.0	6	rev A / list
PDT-204	tx	bed 1 pressure drop	133.0, 131.2	+19.2	+18.0	6	rev A / list
TE/TT-211	tx	bed 2 inlet temperature	128.3, 136.5	+16.1	+15.0	7	rev A / list
TE/TT-212	tx	bed 2 middle temperature	128.3, 138.0	+15.4	+15.0	7	rev A / list
TE/TT-213	tx	bed 2 exit temperature	128.3, 139.5	+14.7	+15.0	7	rev A / list
PDT-214	tx	bed 2 pressure drop	133.0, 131.2	+16.2	+15.0	7	rev A / list
TE/TT-221	tx	bed 3 inlet temperature	128.3, 136.5	+13.3	+12.0	7	rev A / list
TE/TT-222	tx	bed 3 middle temperature	128.3, 138.0	+12.6	+12.0	7	rev A / list
TE/TT-223	tx	bed 3 exit temperature	128.3, 139.5	+11.8	+12.0	7	rev A / list
PDT-224	tx	bed 3 pressure drop	133.0, 131.2	+13.2	+12.0	7	rev A / list
TE/TT-232	tx	bed 4 middle temperature	128.3, 138.0	+9.7	+9.0	8	rev A / list
TE/TT-233	tx	bed 4 exit temperature	128.3, 139.5	+9.0	+9.0	8	rev A / list
PDT-234	tx	bed 4 pressure drop	133.0, 131.2	+10.2	+9.0	8	rev A / list
TE/TT-241	tx	bed 5 inlet temperature	128.3, 136.5	+7.6	+6.0	8	rev A / list
TE/TT-242	tx	bed 5 middle temperature	128.3, 138.0	+6.9	+6.0	8	rev A / list
TE/TT-243	tx	bed 5 exit temperature	128.3, 139.5	+6.2	+6.0	8	rev A / list
PDT-244	tx	bed 5 pressure drop	133.0, 131.2	+7.2	+6.0	8	rev A / list
TE/TT-231	tx	bed 4 inlet (return gas), PV of TIC-231	126.5, 138.0	+10.6	+9.0	8	rev A / list
HV-112	hv	start-up hot-gas line to bed 4, LOCKED CLOSED in operation	124.0, 134.0	+9.6	+9.0	8	rev A / list
ZSC-112	zs	HV-112 closed limit switch (start permissive)	124.5, 133.6	+9.8	+9.0	8	rev A / list
TV-231	cv	SH-201 gas bypass valve position (FO)	128.0, 157.4	+6.8	+6.0	9	rev A / list
TV-211	cv	E-201 hot-side bypass valve position (FO)	140.0, 157.4	+6.8	+6.0	9	rev A / list
TV-221	cv	E-202 hot-side bypass valve position (FO)	150.5, 157.4	+6.8	+6.0	9	rev A / list
TE/TT-207	tx	superheated steam leaving SH-201	124.0, 154.6	+9.0	+6.0	9	rev A / list
TE/TT-253	tx	E-201 hot-side outlet	138.6, 154.5	+3.0	0.0	9	rev A / list
TE/TT-254	tx	return gas leaving E-202	142.0, 151.5	+3.0	0.0	9	rev A / list
TV-251	cv	E-203 gas bypass valve position (FO)	153.5, 132.5	+6.8	+6.0	10	rev A / list
TE/TT-251	tx	gas to IPA T-301	160.0, 134.0	+7.4	+6.0	10	rev A / list
TV-241	cv	E-204 gas bypass valve position (FO)	150.5, 136.6	+7.8	+6.0	10	rev A / list
TE/TT-255	tx	BFW leaving E-203	153.4, 126.0	+2.0	0.0	10	rev A / list
FT-253	tx	BFW to the economizers	151.8, 123.6	+1.5	0.0	10	rev A / list
TE/TT-252	tx	gas to the final absorber T-302	162.5, 128.0	+3.0	0.0	10	rev A / list
P-301A	lcs	pump local control station	171.0, 125.6	+1.2	0.0	11	rev A / list
P-301B	lcs	pump local control station (auto-start HS-301B)	179.0, 125.6	+1.2	0.0	11	rev A / list
HV-311A	hv	P-301A discharge block valve	171.0, 126.2	+6.5	+4.3	11	proposed
HV-311B	hv	P-301B discharge block valve	179.0, 126.2	+6.5	+4.3	11	proposed
PG-311A	pg	P-301A discharge pressure gauge	171.6, 126.8	+5.8	+4.3	11	proposed
PG-311B	pg	P-301B discharge pressure gauge	179.6, 126.8	+5.8	+4.3	11	proposed
FE-301	fe	acid flow element on L-030	175.0, 147.0	+4.5	0.0	11	rev A / list
FT-301	tx	IPA acid circulation (also FSLL-301, I-3)	176.2, 147.0	+1.5	0.0	11	rev A / list
AT-303	tx	IPA acid strength	176.2, 147.8	+1.5	0.0	11	rev A / list

Tag	Kind	What is read or operated there	x, y	EL item	Read from EL	Station	P&ID
TE/TT-304	tx	IPA acid temperature to T-301	175.0, 147.0	+3.0	0.0	11	rev A / list
TE/TT-305	tx	gas leaving T-301	175.8, 140.0	+21.0	+20.2	11	rev A / list
P-302A	lcs	pump local control station	193.0, 125.6	+1.2	0.0	12	rev A / list
P-302B	lcs	pump local control station (manual start)	199.0, 125.6	+1.2	0.0	12	rev A / list
HV-321A	hv	P-302A discharge block valve	193.0, 126.2	+6.5	+4.1	12	proposed
HV-321B	hv	P-302B discharge block valve	199.0, 126.2	+6.5	+4.1	12	proposed
PG-321A	pg	P-302A discharge pressure gauge	193.6, 126.8	+5.8	+4.1	12	proposed
PG-321B	pg	P-302B discharge pressure gauge	199.6, 126.8	+5.8	+4.1	12	proposed
FE-302	fe	acid flow element on L-032	195.0, 147.0	+4.5	0.0	12	rev A / list
FT-302	tx	FAT acid circulation (also FSLL-302, I-4)	196.2, 147.0	+1.5	0.0	12	rev A / list
AT-306	tx	stack SO_2 (CEMS shelter)	209.0, 135.0	+1.5	0.0	12	rev A / list
AT-307	tx	stack acid mist (CEMS shelter)	209.8, 135.0	+1.5	0.0	12	rev A / list
HS-001	hs	ESD push-button (I-6) on the control-room console	60.0, 50.0	+1.0	0.0	CR	rev A / list

Appendix B · Assumptions

- Spacing rules, road and crane distances, bund wall heights, the 2 % acid drain and 1 % sulfur slopes, the 1:200 steam fall: typical practice, assumed.
- Prevailing wind from the SW with the rose frequencies shown (assumed); the stack height 60 m (assumed, above the GEP height).
- Heights: R-201 ring beam EL +4.5, T-301/T-302 skirts to EL +5.0 with T/T 14 m, B-101 drum axis EL +14.0, platforms as listed; pump NPSHr 3.0 m (acid) and 2.5 m (sulfur), submergence 1.0 m (assumed).
- Sizes not in the equipment list: E-201 shell 5 m, E-202 4 m, SH-201 3 m, economizer casings 4 × 6 m, TK-301 3.8 × 12.4 m, TK-302 3.6 × 9 m, filters Ø 1.6 m, coolers 8 × 4 m (assumed).
- Proposed tags (not on rev A): HV-113A, HV-113B, PG-113A, PG-113B, HV-114A, HV-114B, PDG-114, HV-108, LG-108, HV-311A, HV-311B, PG-311A, PG-311B, HV-321A, HV-321B, PG-321A, PG-321B (query EQ-la-01).
- Materials on the isometrics: typical for the service; isometric lengths are routing lengths of this teaching layout, not fabrication data.